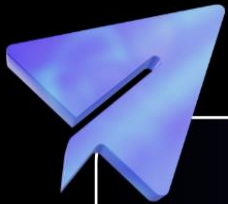


CSE485: Deep Learning

Tennis Object Detection



**Presented to Dr. Alaa Hamdy
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Real-Time Tennis Object Detection Using YOLOv8

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Abstract — This paper presents a deep learning-based system for real-time detection of tennis objects—specifically tennis balls, rackets, and players—in broadcast video footage using the YOLOv8 architecture. We address a key practical challenge in sports computer vision: the aggregation and harmonization of heterogeneous public datasets with inconsistent class naming conventions. Six datasets sourced from Roboflow Universe were merged through an automated label-remapping pipeline, yielding approximately 2,650 images after deduplication and an 80/10/10 train/val/test split. We conduct an ablation study across four model configurations comparing YOLOv8n, YOLOv8s, and YOLOv8m, with systematic dataset expansion. Our final model—YOLOv8s trained for 100 epochs on the merged six-dataset corpus—achieves an overall mAP@50 of 0.809 on the validation set and 0.820 on the held-out test set, with per-class mAP@50 of 0.905 (ball), 0.527 (racket), and 0.995 (player). The racket class presents a persistent challenge attributed to dataset imbalance and visual occlusion. We further analyze distribution shift as an explanation for reduced player detection confidence in close-up video footage unseen during training. **GitHub repository link:** <https://github.com/carolrb-123/CSE485-Deep-Learning--Tennis-Object-Detection>

Video Tests: <https://drive.google.com/drive/folders/1pyxBnsYkXFPUEmz6TV8d6AkddHGpg8is?usp=sharing>

Keywords—Object detection, YOLOv8, tennis, sports analytics, transfer learning, dataset merging, real-time inference.

I. INTRODUCTION

Automated sports analysis is an increasingly active research area, driven by demand from broadcasters, coaching staff, and analytics platforms. Tennis, in particular, poses several unique challenges for computer vision systems: the ball is small, fast-moving, and frequently motion-blurred; rackets undergo rapid orientation changes; and players appear at widely varying scales depending on camera position.

Existing work has addressed individual aspects of this problem—dedicated ball trackers, pose estimators for players, or court detection systems—but few unified multi-class detection pipelines are publicly available at low cost. This work aims to bridge that gap by building a single YOLOv8-based detector capable of simultaneously localizing all three primary tennis objects in broadcast-quality footage.

A significant practical contribution of this work is the dataset aggregation methodology. Public datasets on Roboflow Universe use inconsistent class nomenclature (e.g., "Ball", "tennis ball", "ball") which, if naively merged, causes severe label noise. We present an automated remapping pipeline that resolves this issue and enables principled multi-source training.

The remainder of this paper is organized as follows: Section II reviews related work; Section III describes the dataset and preprocessing pipeline; Section IV details the model architecture and training procedure; Section V presents experimental results and ablation analysis; Section VI discusses limitations and future work; and Section VII concludes.

II. RELATED WORK

A. Object Detection Architectures

The YOLO (You Only Look Once) family of models has dominated real-time object detection since its introduction by Redmon et al. [1]. YOLOv8, released by Ultralytics in 2023, introduces anchor-free detection, a decoupled head, and improved data augmentation, achieving state-of-the-art speed-accuracy tradeoffs on COCO [2]. Its pretrained backbone (trained on COCO) enables effective transfer learning even with small domain-specific datasets.

B. Tennis-Specific Detection

Wang et al. [3] proposed YOLOBR combined with FTOC tracking for robust tennis ball detection in complex video streams. Their system was trained on a private dataset of 15 video streams collected in-house, highlighting the data scarcity problem for tennis-specific training. TrackNet [4] introduced a specialized deep learning approach for tennis ball tracking using heatmap regression, achieving high accuracy on small fast-moving balls but requiring video sequences rather than single-frame inference. The MATLAB Deep Learning Blog [5] demonstrated a ball tracking pipeline for tennis analysis using transfer learning, emphasizing the utility of pretrained feature extractors for domain adaptation.

C. Dataset Aggregation

Multi-source dataset merging is an underexplored problem in applied computer vision. Inconsistent ontologies across datasets introduce label noise that degrades detection performance [6]. Our work contributes a systematic remapping approach applicable to any multi-source Roboflow-exported dataset.

III. DATASET AND PREPROCESSING

A. Dataset Collection

Six publicly available datasets were sourced from Roboflow Universe. Table I summarizes each dataset. A critical observation is that the same physical object (e.g., a tennis ball) is labeled under multiple class names across datasets, including "Ball", "tennis ball", "ball", "0", and others.

Dataset	Images	Classes
Tennis Object Detection with YOLO	416	Ball, Player, Racquet
Tennis Ball Detection	245	tennis ball
My First Project (co-labeled)	39	Ball, Player, Racket
Racket (v1)	~200	Racket
Tennis Racket (Hoo)	~300	Racket, ball
Tennis Racket (Hyundong)	~300	Racket, ball

Table I-Dataset Summary

B. Label Remapping Pipeline

To address class name inconsistency, we developed an automated Python pipeline that: (1) reads each dataset's *data.yaml* to extract class names; (2) applies a hand-verified alias dictionary mapping all variant names to three canonical classes (*tennis_ball*, *tennis_racket*, *tennis_player*); (3) rewrites all YOLO-format annotation files with corrected class indices; and (4) copies images and remapped labels into a unified output directory. Annotations belonging to excluded classes (e.g., "object", "overlap") are silently dropped. Filename collisions across datasets are resolved by prefixing each file with a dataset index.

The alias dictionary was constructed conservatively: ambiguous class names such as "person" and "object" were inspected visually on a per-dataset basis before mapping. In one dataset, the class "object" was found to annotate miscellaneous content (sky, full-frame crops) across only 7 images and was dropped entirely.

C. Train/Validation/Test Split

All source datasets were delivered without consistent splits—several had no validation folder at all. After merging, the full image pool was globally shuffled (random seed = 42) and partitioned 80/10/10 into training, validation, and test sets. This strategy prevents data leakage between splits and ensures all three sets share the same distribution.

IV. MODEL ARCHITECTURE AND TRAINING

A. Architecture

We use YOLOv8s (small variant) as our final model. YOLOv8s comprises 73 layers and 11.1 million parameters, with 28.4 GFLOPs of computation per inference pass at 640×640 input resolution. The architecture uses a CSPDarknet backbone, a PANet feature pyramid neck, and an anchor-free detection head with decoupled classification and regression branches. All experiments are initialized from COCO-pretrained weights (*yolo8s.pt*), enabling transfer learning of low-level and mid-level visual features.

B. Training Configuration

Training was performed on Google Colab using an NVIDIA Tesla T4 GPU (14.9 GB VRAM). The final training configuration is as follows: 100 epochs, image resolution 640×640, batch size 8, AdamW optimizer with initial learning rate 0.0005 and cosine learning rate decay, early stopping with patience 30, and label smoothing 0.1. The following data augmentation strategies were applied: horizontal flip ($p=0.5$), mosaic augmentation (scale 1.0), MixUp ($\alpha=0.1$), HSV color jitter ($H=0.015$, $S=0.7$, $V=0.4$), random scale ($\pm 50\%$), translation ($\pm 10\%$), and rotation ($\pm 10^\circ$).

Mosaic augmentation is particularly beneficial in this setting as it synthesizes composite training images from four randomly selected samples, implicitly creating multi-object scenes that complement the many single-class images in the ball-only datasets.

V. EXPERIMENTS AND RESULTS

A. Ablation Study

Four model configurations were evaluated to isolate the contribution of dataset expansion and model capacity. Results are reported as per-class mAP@50 on the validation set. Table II summarizes the ablation.

Experiment	Model	Datasets	Ball	Racket	Player
Baseline	YOLOv8s	2	0.671	0.539	0.992
+ Co-labeled data	YOLOv8s	3	0.716	0.473	0.994
Larger model	YOLOv8m	3	0.758	0.548	0.994
Final (6 datasets)	YOLOv8s	6	0.905	0.527	0.995

Table II-Ablation Study: mAP@50 per Class

Several observations emerge from the ablation. First, adding the 39-image co-labeled dataset (Experiment 2) improved ball detection but degraded racket performance, likely because the small dataset introduced distribution mismatch without sufficient racket diversity. Second, upgrading to YOLOv8m (Experiment 3) improved all metrics, confirming the benefit of increased capacity when co-occurrence data is available. Third, expanding to six datasets (Final) yielded the largest single improvement in ball detection (0.905 mAP@50), confirming that data volume is the primary driver for this class, while racket accuracy remained constrained by class imbalance.

B. Final Model Test Results

Table III reports test-set performance of the final YOLOv8s model. The test set consists of 294 images with 459 total annotations across the three classes.

Class	Images	Instances	Precision	Recall	mAP@50
All	294	459	0.929	0.773	0.809
tennis_ball	243	243	0.922	0.877	0.905
tennis_racket	67	106	0.882	0.453	0.527
tennis_player	61	110	0.982	0.989	0.995

Table III-Test Set Detection Results — Final Model (YOLOv8s, 6 Datasets)

C. Training Curves

Fig. 1 shows the training and validation loss curves alongside precision, recall, and mAP metrics over 100 epochs. All three loss components (box, classification, DFL) decrease monotonically for both train and validation splits, indicating no overfitting. The validation mAP@50 curve rises steeply in the first 40 epochs and plateaus near 0.82, consistent with the quantitative results.

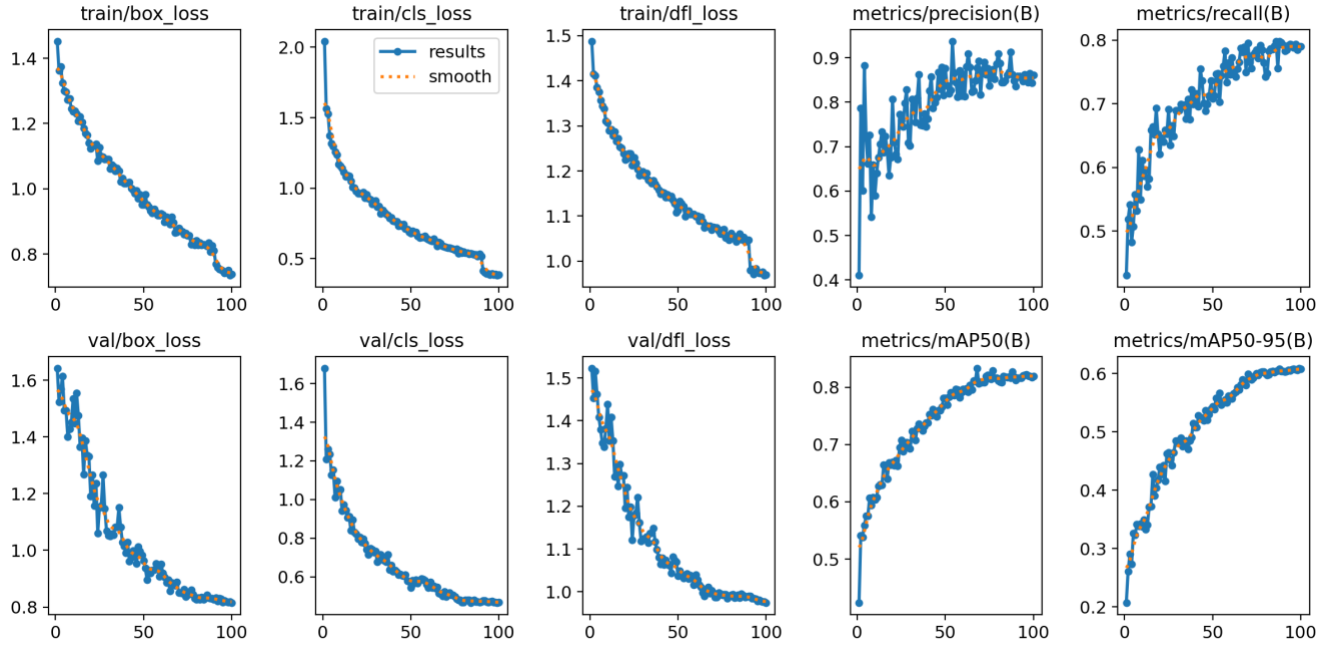


Figure 1-. Training and validation loss curves, precision, recall, and mAP@50 over 100 epochs.

D. Precision-Recall Analysis

Fig. 2 presents the Precision-Recall curve per class. The tennis_player class achieves near-perfect area under the curve (0.995 mAP@50). The tennis_ball class reaches 0.919 AP@50, with precision remaining high (>0.85) up to recall ≈ 0.8 before degrading. The tennis_racket class shows substantially lower area (0.545 AP@50) and a steeply declining precision-recall curve, reflecting the difficulty of detecting rackets under occlusion and class imbalance.

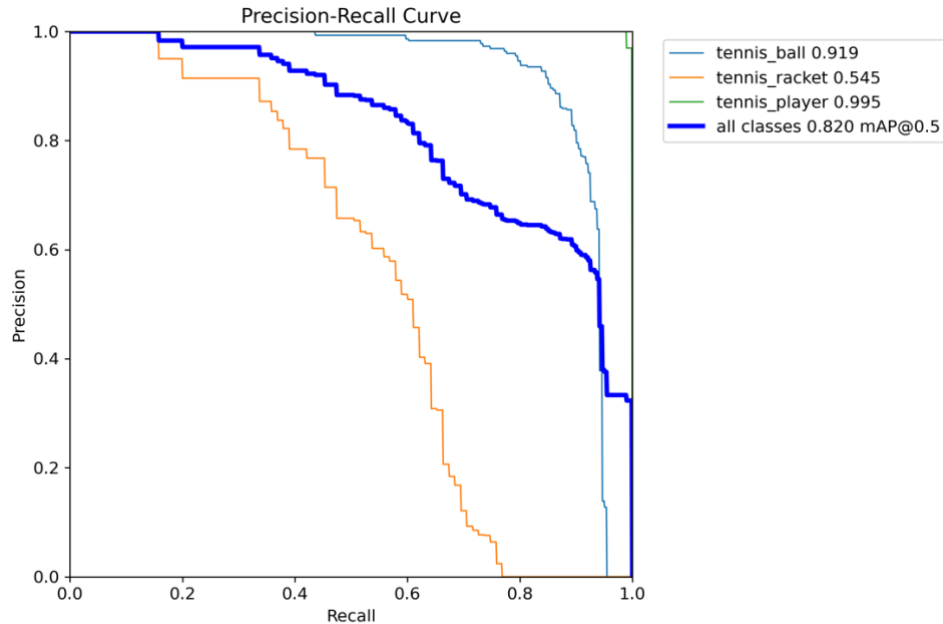


Figure 2-Precision-Recall curve per class. Overall mAP@50 = 0.820.

E. F1-Confidence Analysis

Fig. 3 shows the F1-Confidence curve. The optimal confidence threshold for all classes combined is 0.383, yielding $F1 = 0.80$. The tennis_player class maintains $F1 > 0.90$ across a wide confidence range (0.05–0.92), while tennis_racket peaks at $F1 \approx 0.58$ and drops off more steeply. This informs the choice of confidence threshold for deployment: $\text{conf}=0.3$ is used during inference to maximize racket recall at acceptable precision.

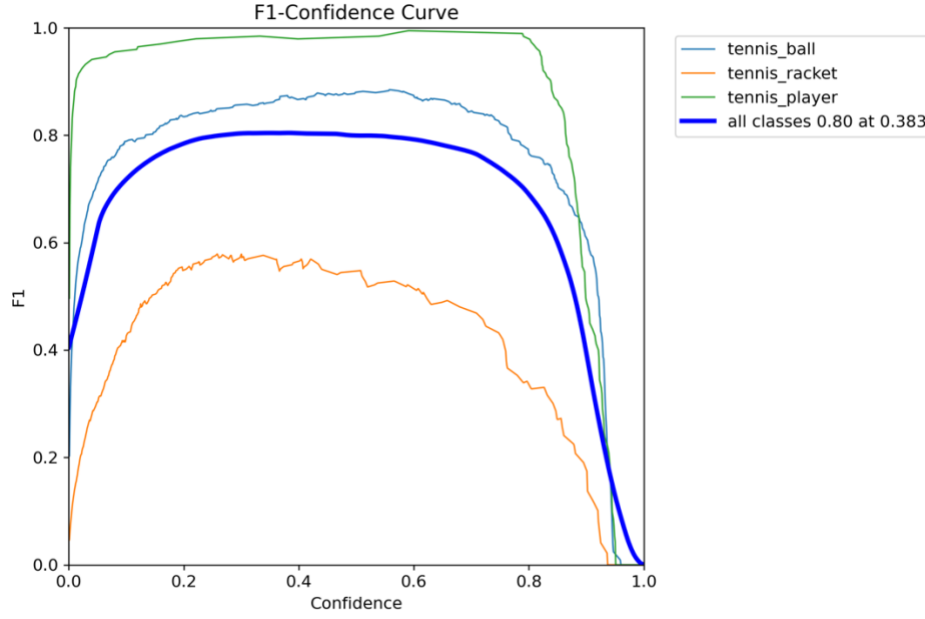


Figure 3-F1-Confidence curve per class. Optimal threshold: 0.383 (all classes $F1 = 0.80$).

F. Confusion Matrix

Fig. 4 shows the normalized confusion matrix on the test set. The model correctly classifies 90% of ball instances and 100% of player instances. Racket recall is 60%, with 40% of racket instances classified as background—the primary failure mode. This is consistent with the low recall (0.453) reported in Table III and confirms that missed detections rather than misclassifications are the dominant error type for rackets.

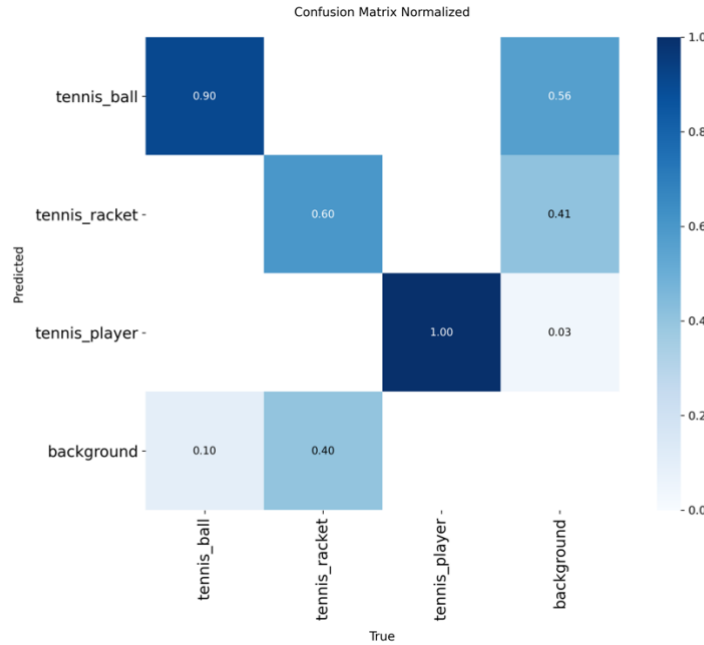


Figure 4-Normalized confusion matrix on the test set.

G. Qualitative Results

Figs. 5 and 6 show ground-truth labels and model predictions on two validation batches. The model accurately localizes balls in broadcast footage at varying distances, correctly identifies players across multiple court types (clay, grass, hard court), and successfully detects rackets in close-up action images with high confidence (0.9).

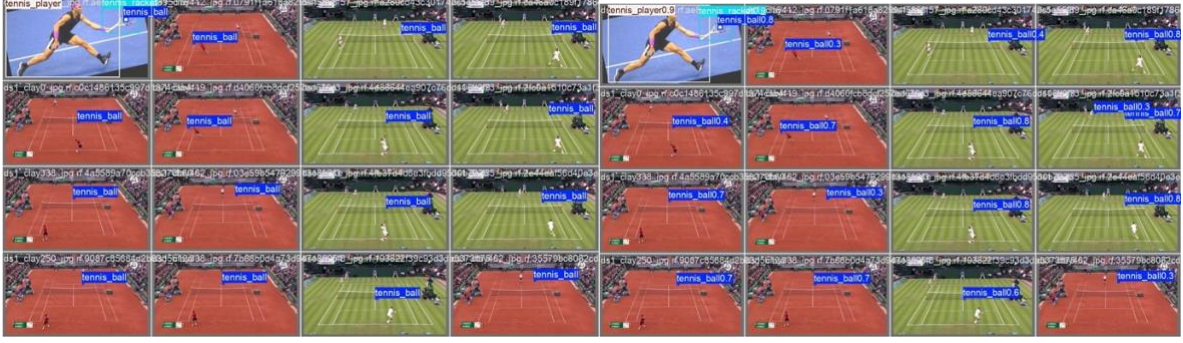


Figure 5-Left: Ground-truth annotations. Right: Model predictions (val batch 0).



Figure 6-Left: Ground-truth annotations. Right: Model predictions (val batch 1).

H. Video Inference

Figs. 7–9 show sample frames from live inference on broadcast tennis footage. The model operates at approximately 23 ms per frame on the T4 GPU (~43 FPS), well above real-time requirements. Ball detection is consistent across varying court colors and lighting conditions. Player detection confidence degrades in close-up non-broadcast footage (discussed in Section VI).



Figure 7-Video inference: all three classes detected in a close-up frame (player=0.69, racket=0.48, ball=0.87).



Figure 8-Video inference: player detected at medium confidence (0.32) in a close-up frame—illustrating distribution shift from training data.



Figure 9-Video inference: player (0.84) and racket (0.46) correctly detected in a court-side frame.

VI. DISCUSSION

A. Racket Detection Limitations

Tennis racket detection remains the primary performance bottleneck, with $mAP@50$ of 0.527 and recall of 0.453. The confusion matrix confirms that 40% of racket instances are missed entirely rather than misclassified. Three factors contribute to this: (1) class imbalance—the training corpus contains substantially fewer racket annotations than ball annotations; (2) visual occlusion—rackets are frequently partially obscured by the player's hand, arm, or body; and (3) rapid motion blur during swings, which deforms the racket's visual appearance. Future work should address these via targeted data collection focusing on in-swing frames, and potentially a separate racket-specific fine-tuning stage.

B. Distribution Shift in Video Inference

Player detection confidence degrades markedly in close-up technique footage compared to broadcast match footage. This is a classical instance of distribution shift: the model's training data consists predominantly of overhead broadcast images where players occupy a small fraction of the frame, appear full-body, and are viewed from a standardized camera angle. Close-up footage presents players waist-up, filling the majority of the frame, with a non-standard background. Despite these conditions, the model still produces detections (confidence 0.32–0.84), demonstrating some generalization. Addressing this limitation would require augmenting the training set with diverse camera angles.

C. Dataset Merging Methodology

The label-remapping pipeline developed in this work is a reusable contribution for any project aggregating heterogeneous YOLO-format datasets. The key design decisions—conservative alias mapping, per-dataset class inspection, global reshuffle before splitting—prevent the silent failures (label noise, split contamination) that commonly afflict naive dataset concatenation. The pipeline correctly processed all six datasets with zero annotation errors detected in qualitative inspection.

VII. CONCLUSION

We presented a real-time tennis object detection system based on YOLOv8s, trained on a harmonized corpus of six public datasets comprising approximately 2,650 images. Our automated label-remapping pipeline resolves class naming inconsistencies across heterogeneous Roboflow exports, enabling principled multi-source training. The final model achieves an overall mAP@50 of 0.809 on validation and 0.820 on the held-out test set, with near-perfect player detection (0.995) and strong ball detection (0.905). Racket detection (0.527) is identified as the primary limitation, attributed to class imbalance and occlusion. The system operates at approximately 43 FPS on an NVIDIA Tesla T4, satisfying real-time deployment requirements. The ablation study demonstrates that data volume is the dominant performance driver at this scale, and that transfer learning from COCO-pretrained weights enables competitive performance with fewer than 3,000 training images.

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